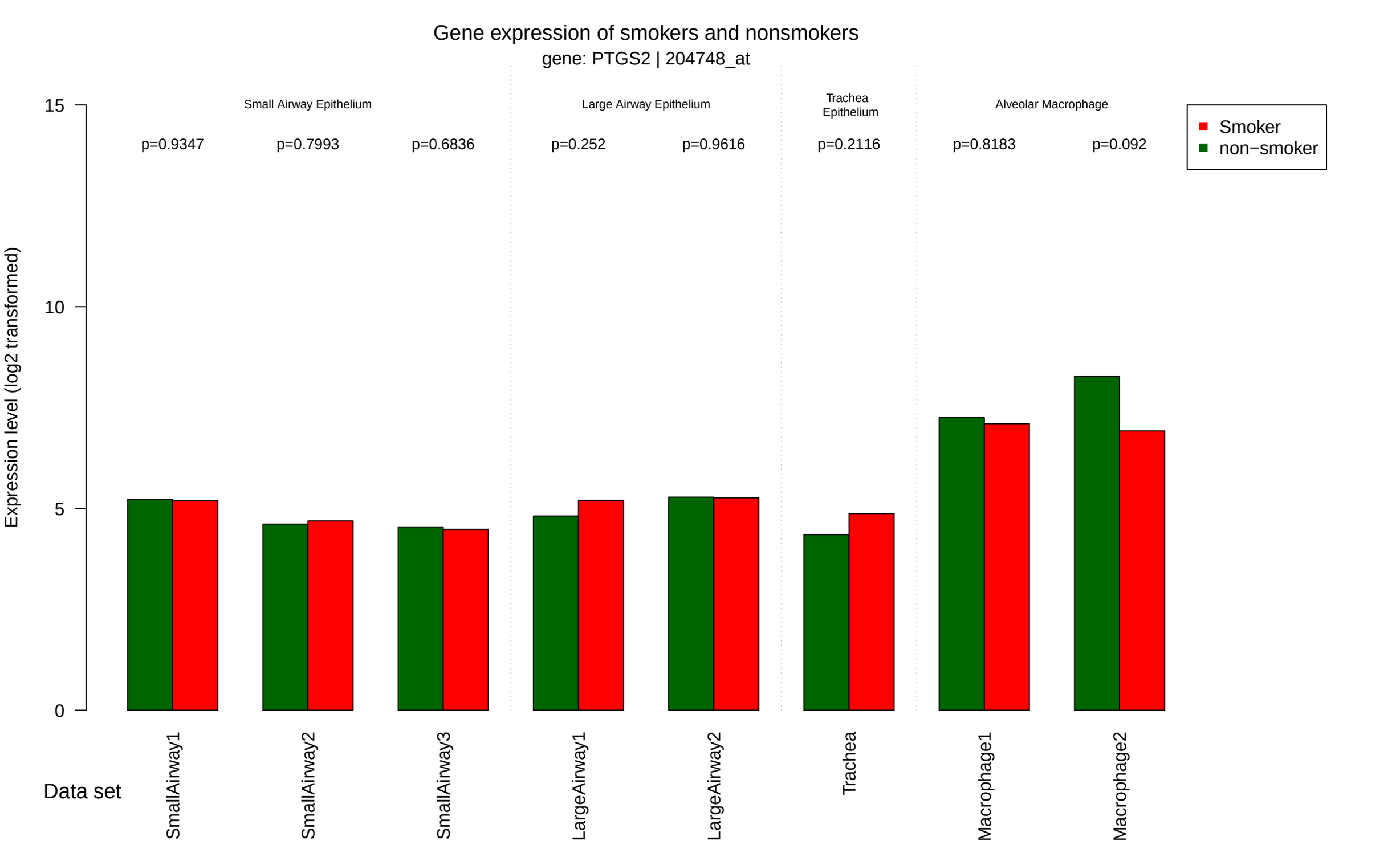
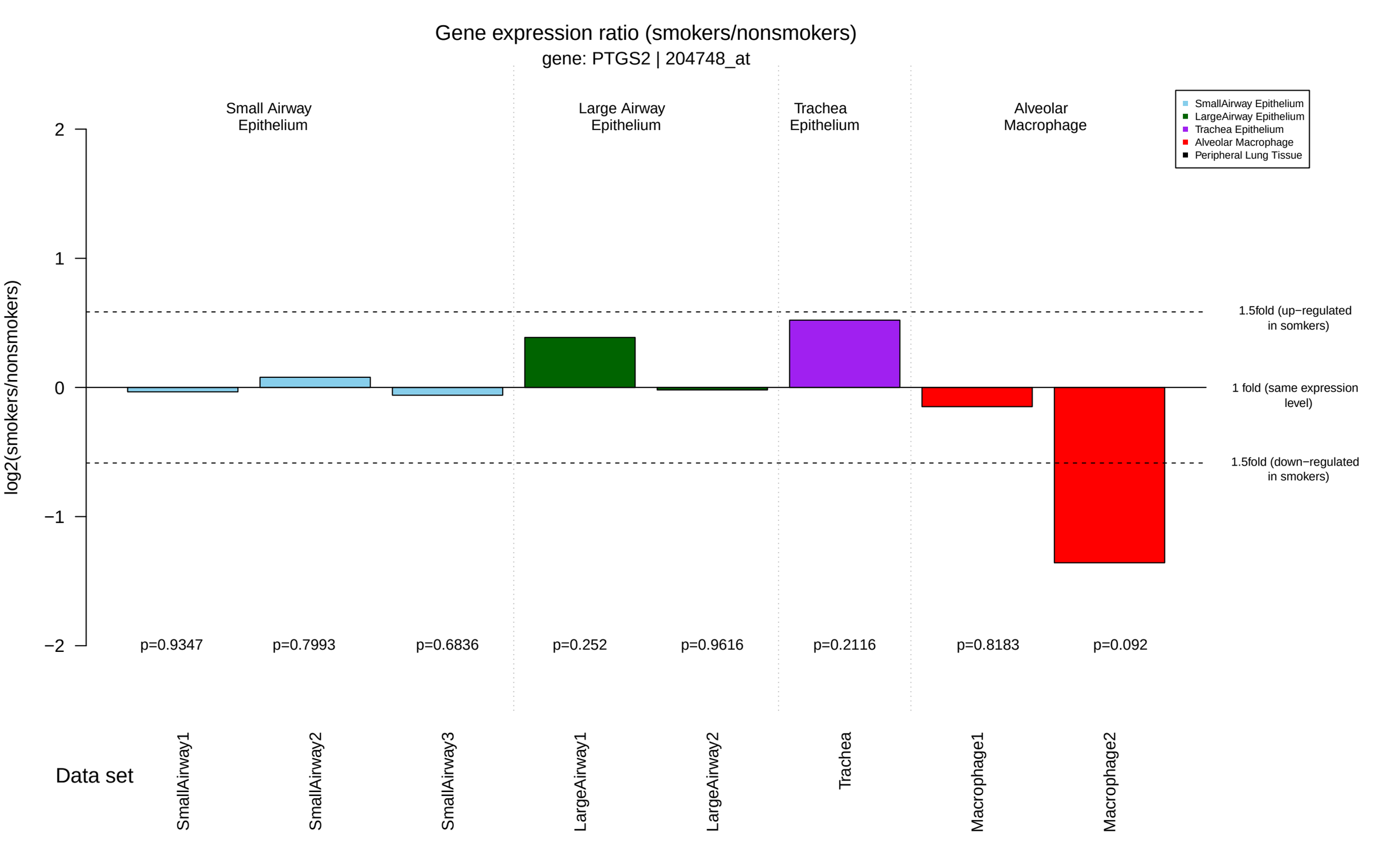
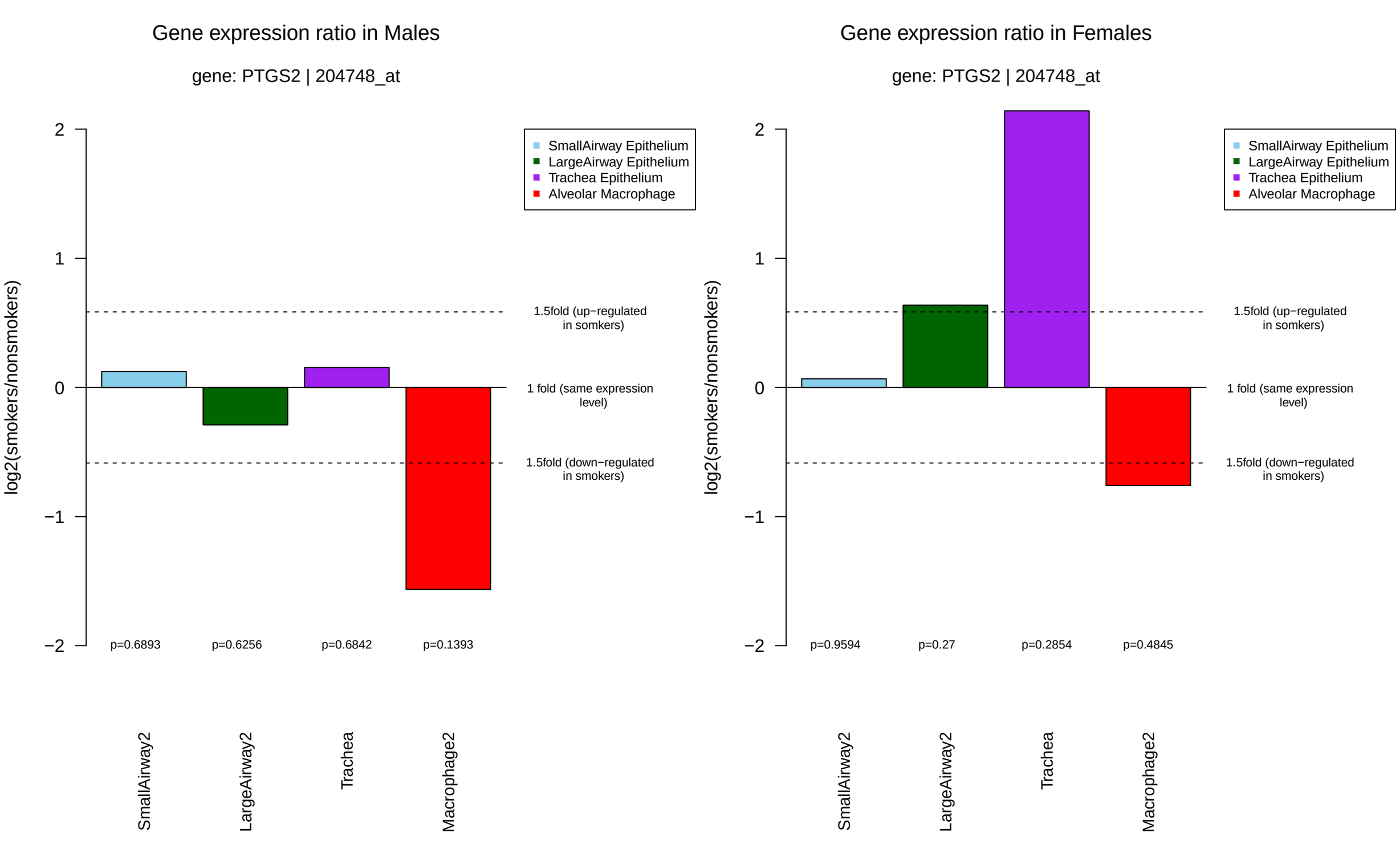
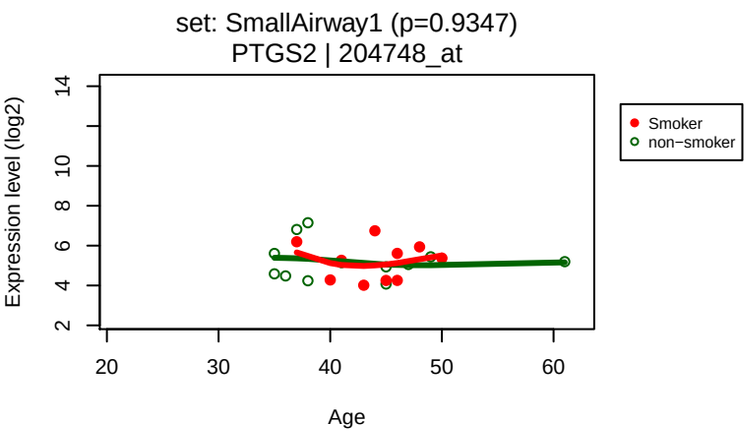
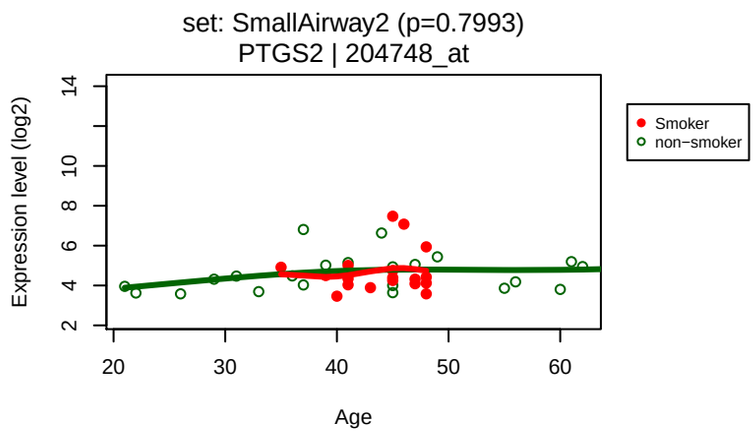
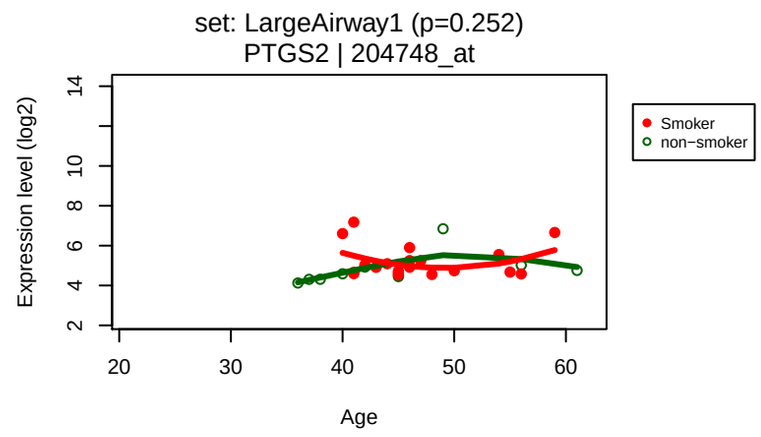

Gene expression of smokers and nonsmokers
gene: PTGS2 | 204748_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9347
p=0.7993
p=0.6836
p=0.252
p=0.9616
p=0.2116
p=0.8183
p=0.092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

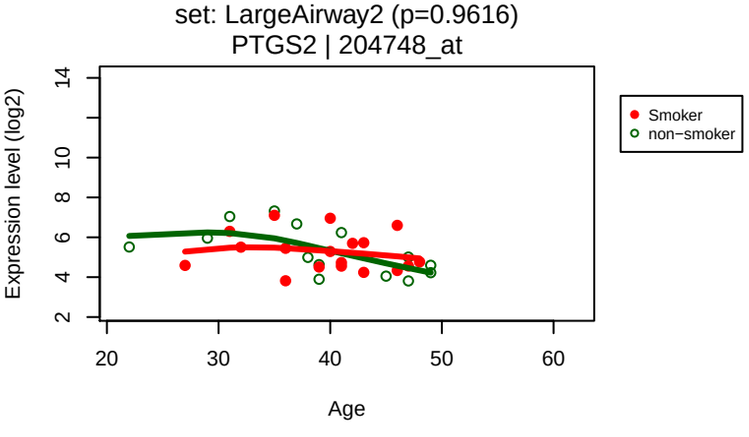

Gene expression ratio (smokers/nonsmokers)
gene: PTGS2 | 204748_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9347
p=0.7993
p=0.6836
p=0.252
p=0.9616
p=0.2116
p=0.8183
p=0.092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

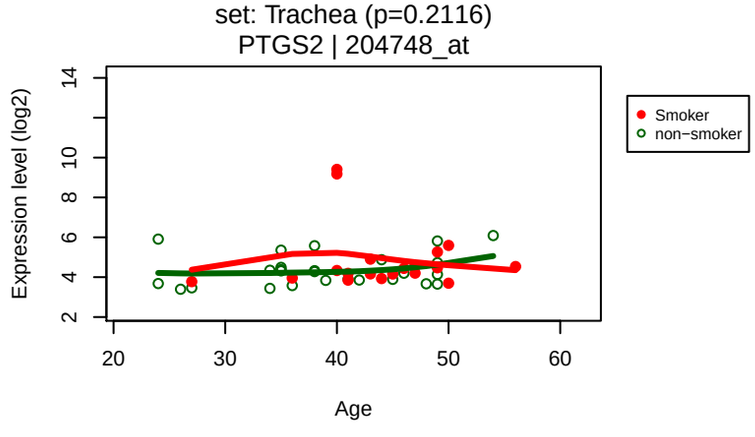

Gene expression ratio in Males
gene: PTGS2 | 204748_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6893
p=0.6256
p=0.6842
p=0.1393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGS2 | 204748_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9594
p=0.27
p=0.2854
p=0.4845
SmallAirway2
LargeAirway2
Trachea
Macrophage2

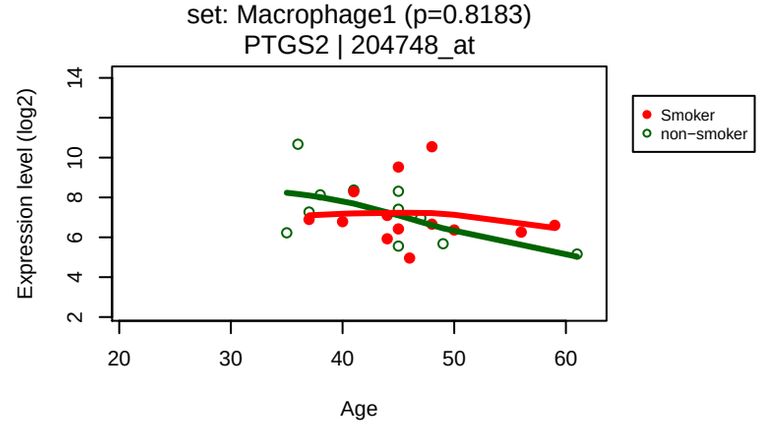

set: SmallAirway1 (p=0.9347)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker

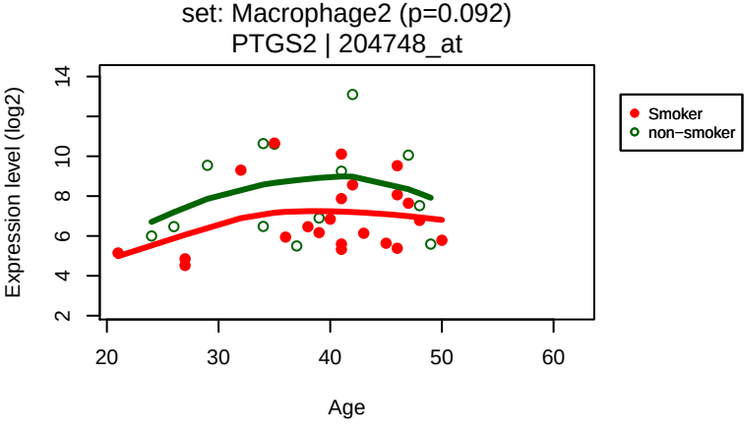

set: SmallAirway2 (p=0.7993)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker

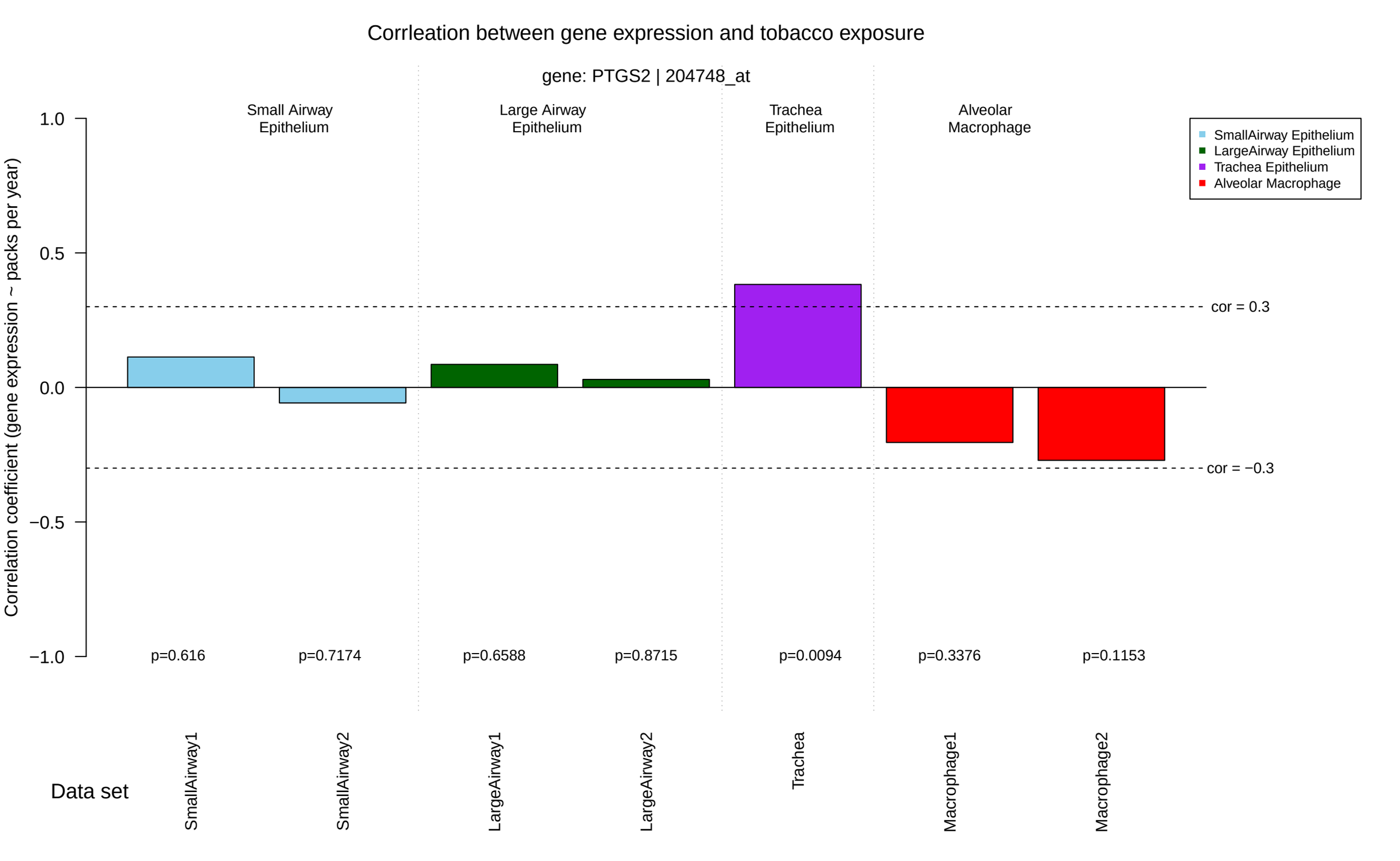

set: LargeAirway1 (p=0.252)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9616)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2116)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8183)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.092)
PTGS2 | 204748_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGS2 | 204748_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.616
p=0.7174
p=0.6588
p=0.8715
p=0.0094
p=0.3376
p=0.1153
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2